

Predicting Breast Cancer Survival

Five-year Survival Prediction Of The Breast Cancer Patients From UK & Canada
Based On The Clinical Information

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Introduction

Context & Clinical motivation:

- Breast cancer is among the leading cause of death
- Outcome of treatments become important for customized therapeutic and improving patient outcomes

Questions:

- A. How does the survival differ between different breast cancer subtypes?
- B. Can we predict 5-year survival of the breast cancer patients given clinical information?



Introduction

Dataset:

- Patient demographics
- Pathology
- Tumor grades
- Receptor classifications

Novelty:

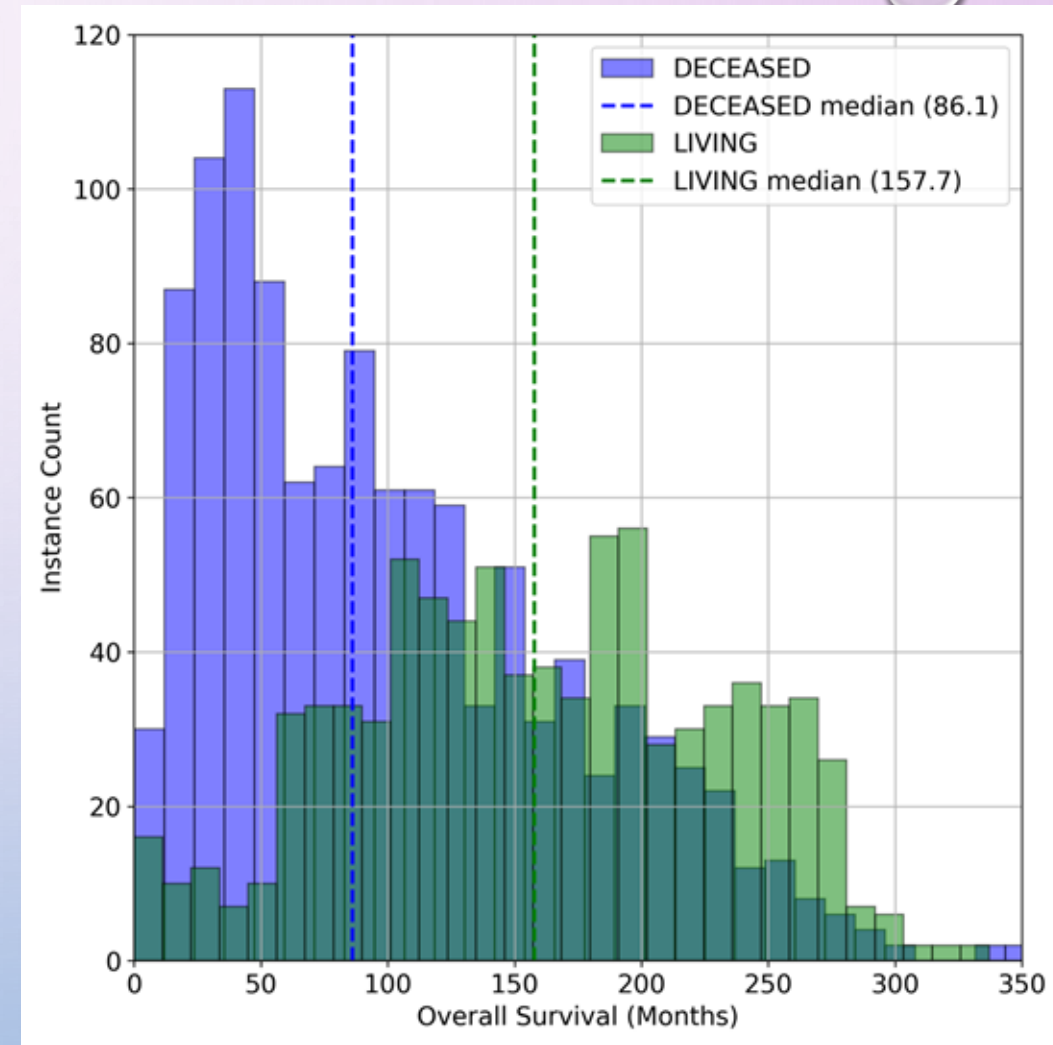
- Clinically relevant way of handling missing data
- Integrating genomic information to clinical features for model improvement
- Versatile model to predict survival of patients from different hospitals

KPIs:

- Main KPIs: ROC-AUC, PR-AUC, MCC and F1
- Precision and Recall

Stakeholders:

- Patients
- Researchers, Hospitals & Pharma companies
- Insurance companies



Initial Data Audit

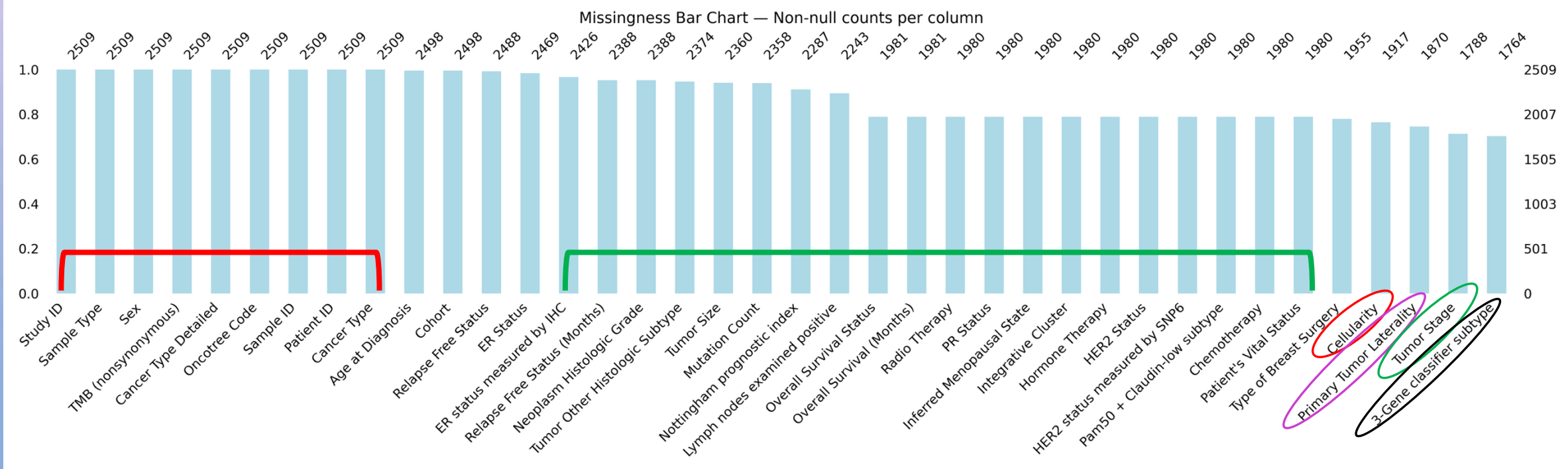
Cohort Completeness Overview: profiles data density across all clinical, and demographic features for 2,509 patients.

100% Complete Fields: Demographics: Study Id, Sex, & Age

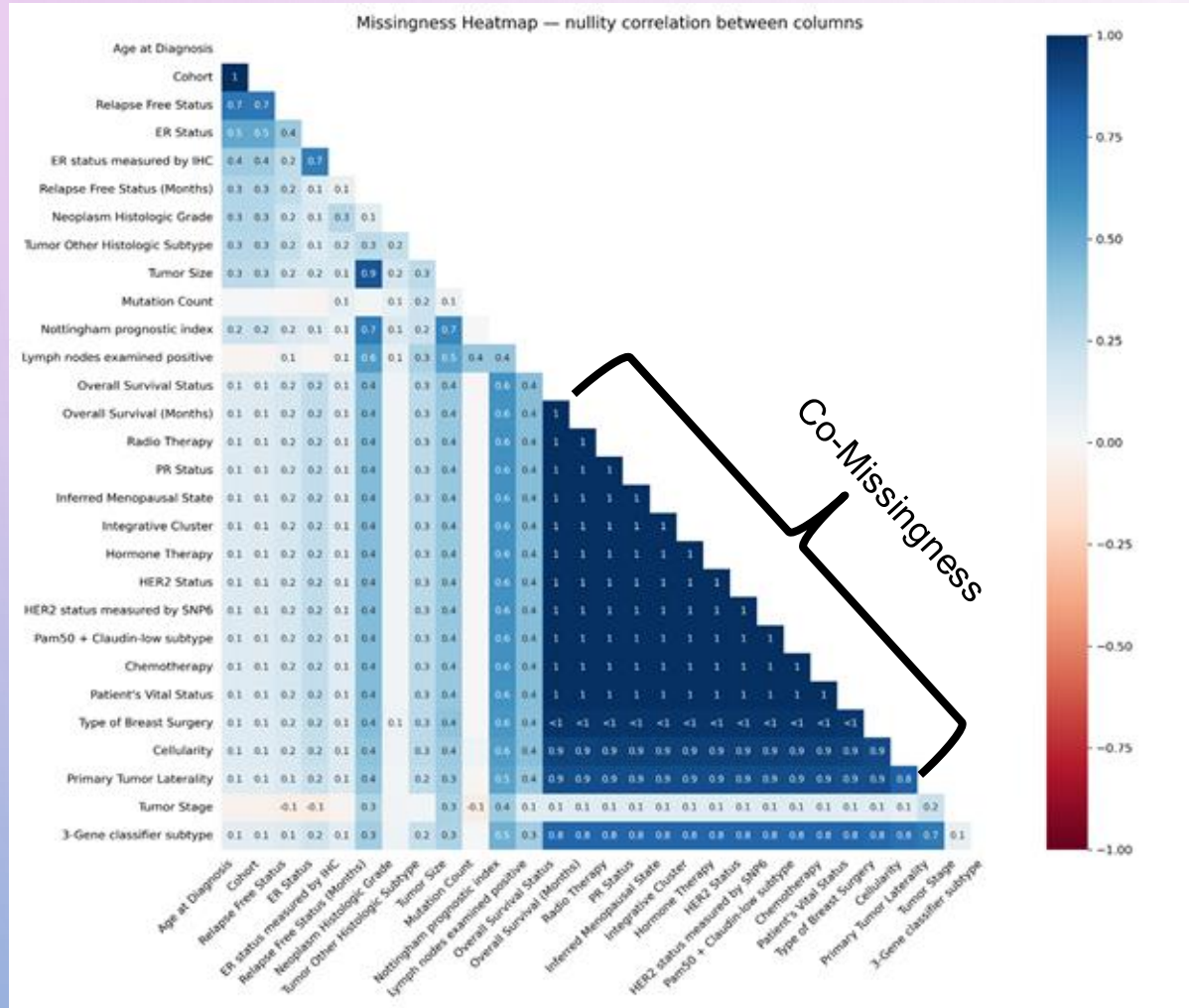
Clinical Variables: Clinical and Therapy variables stabilize at exactly 1,980 non-null records (~79% density).

Maximum Sparsity Risk: Cellularity, Primary Tumor Laterality, Tumor Stage, and 3-Gene classifier subtype drop to 1,764 records (~70% dense).

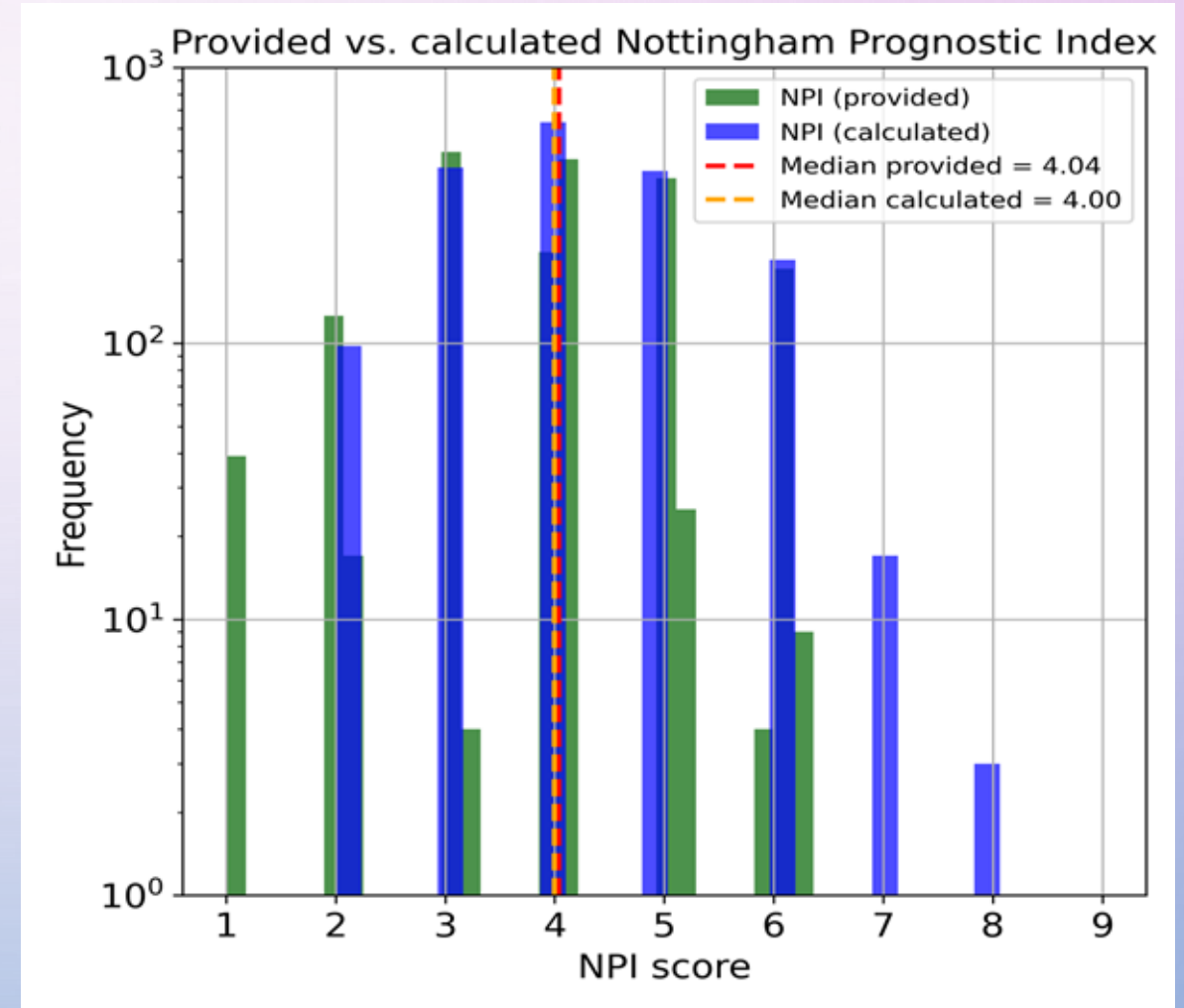
Imputation Justification: High scattered missingness rules out row deletion



Initial Data Audit



Nullity Correlation: Absence of data in features is correlated

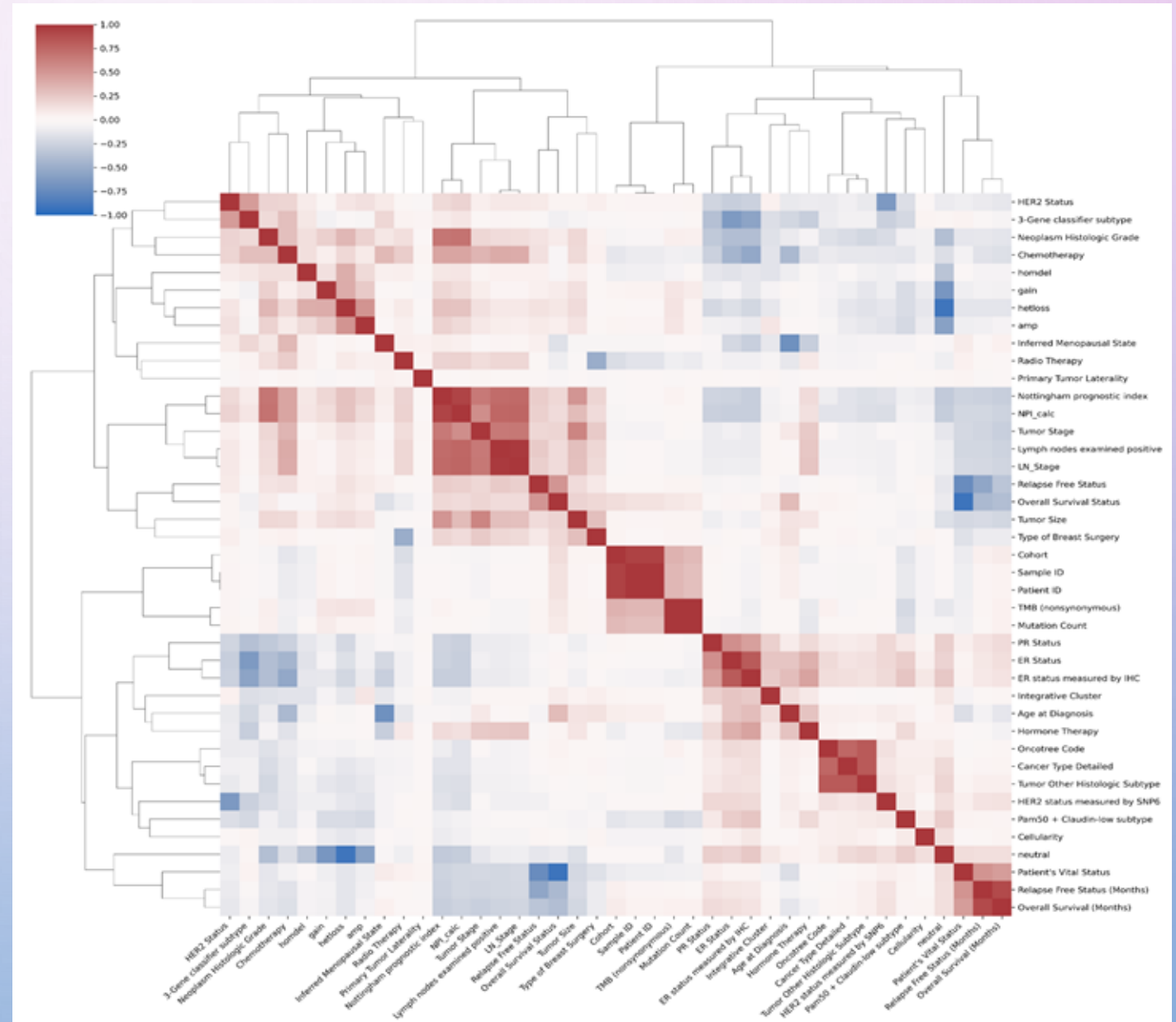


NPI
 $= 0.2 \times \text{tumor diameter} + \text{histological grade (1-3)}$
 $+ \text{lymph node stage (1-3)}$

Feature Correlation & Clustering

Multivariate Clusters: Hierarchical clustering features into modules based on correlation, exposing biological and clinical relationships.

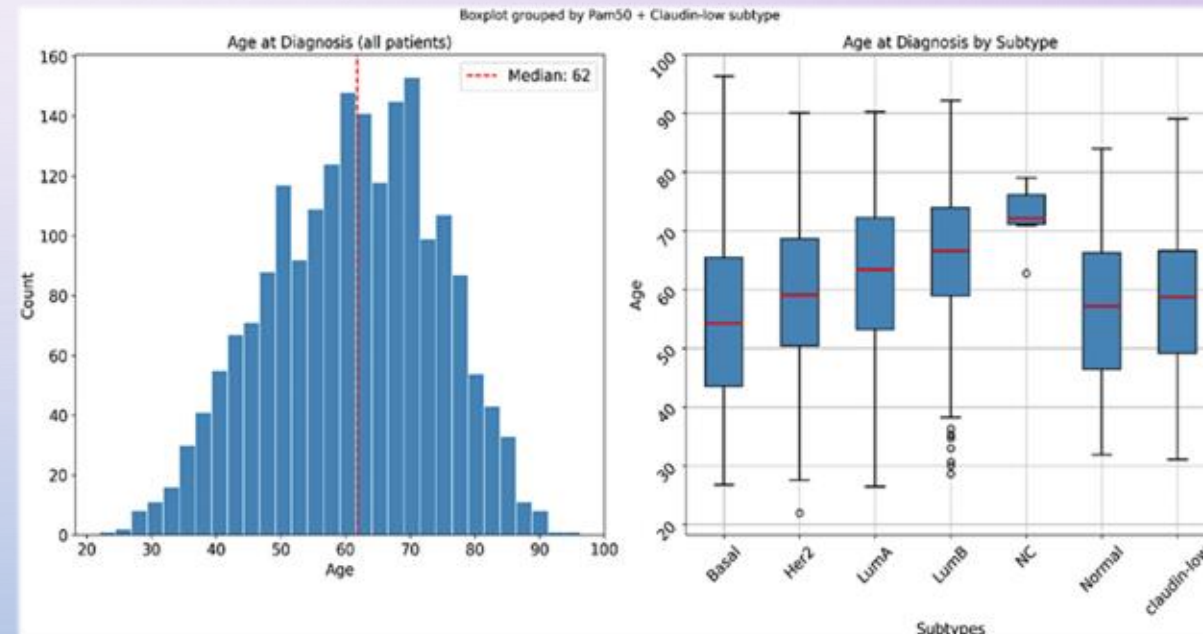
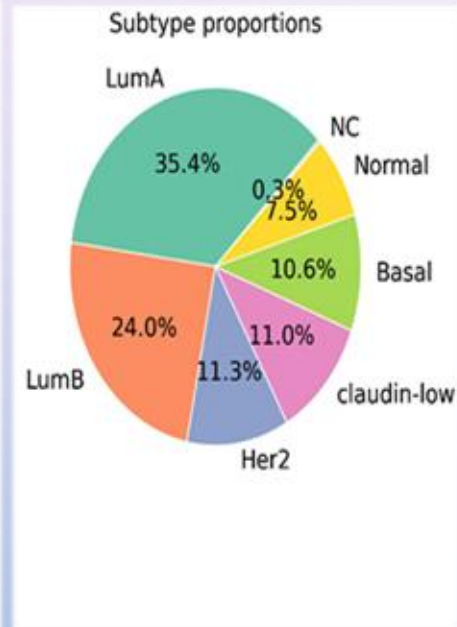
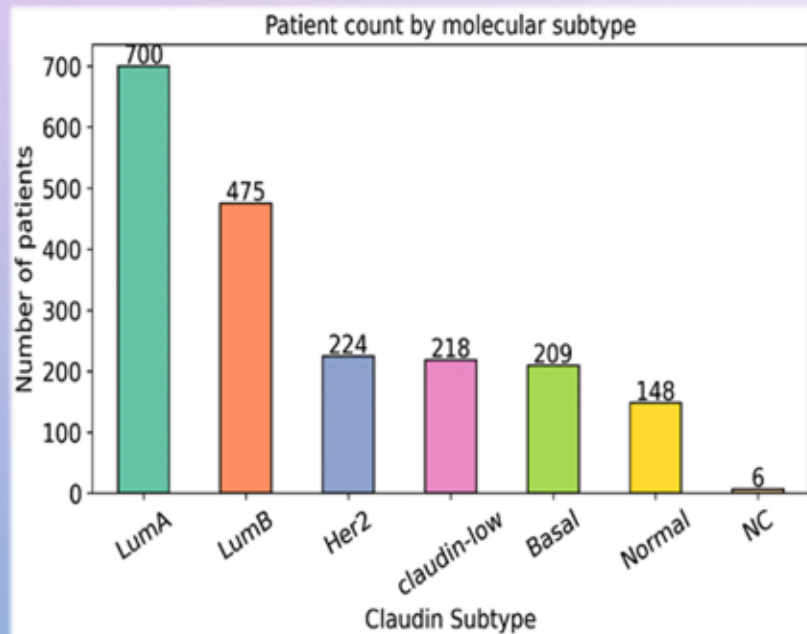
1. **Enzyme block:** PR Status, ER Status, ER status measured by IHC
2. **Cancer Type block:** Oncotree Code, Cancer Type Detailed, Tumor Other Histologic Subtype
3. **NPI block:** Nottingham prognostic index, NPI_calc, LN_Stage, Lymph nodes examined positive, Neoplasm Histologic Grade, Tumor Stage
4. **Mutation block:** Mutation Count, TMB (nonsynonymous)
5. **HER2, 3-gene block:** 3-Gene classifier subtype, HER2 Status
6. **CNA blocks:** homdel, hetloss, neutral, gain, amp



COHORT COMPOSITION BY SUBTYPE

Problem Statement 1:

How does survival differ across breast cancer subtypes — and how much worse off are triple-negative patients compared to other cancer subtypes?

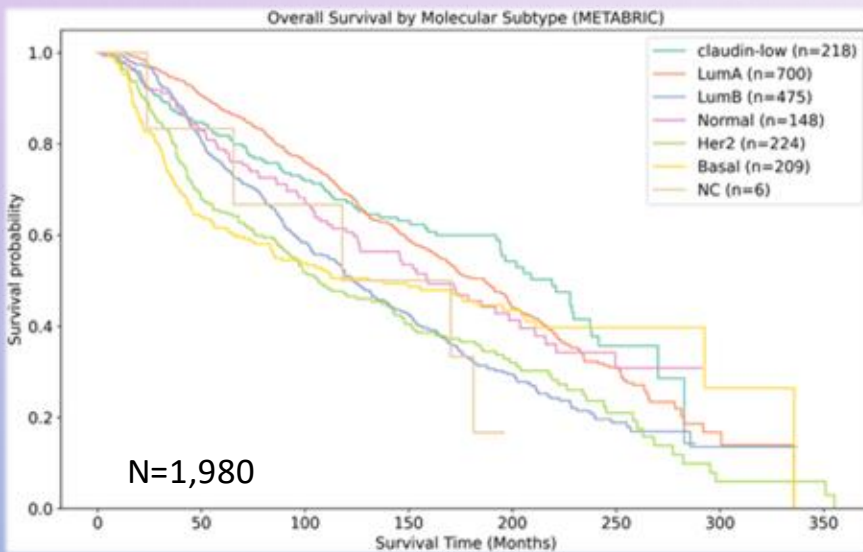


Subtype mix: LumA (~38%) & LumB (24%) dominate;
Basal (10.6%) & Claudin-low (11%) are the TNBC-risk groups

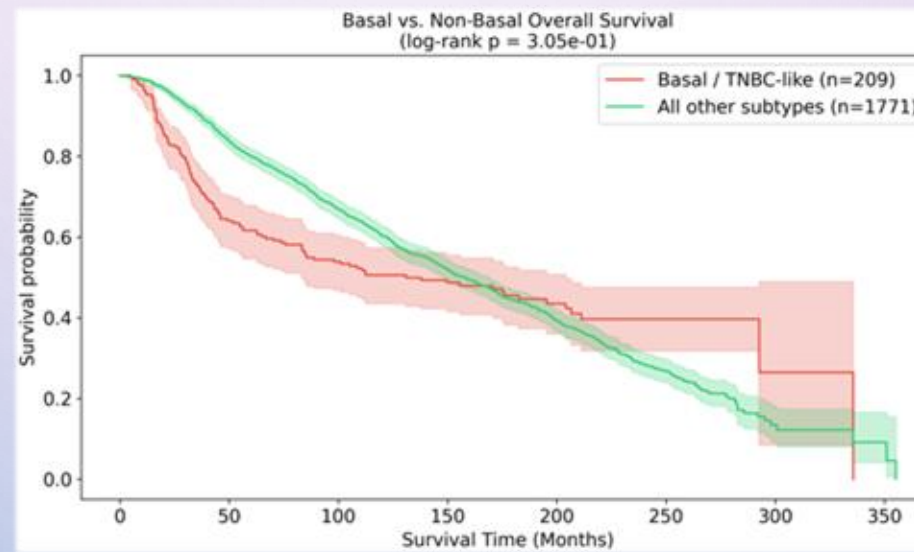
Age at diagnosis: median 62; Basal & Claudin-low skew
younger — consistent with known TNBC biology

SURVIVAL DIFFERS SHARPLY BY SUBTYPE

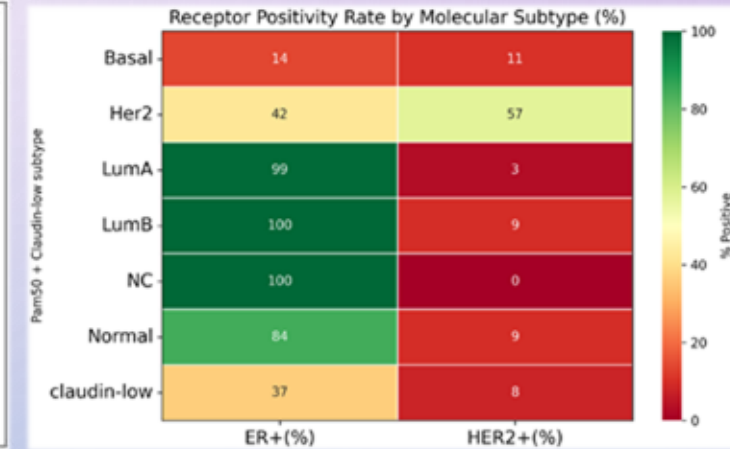
Basal/TNBC-like tumors have the poorest early survival — confirmed by the receptor profile



*KM curves diverge early —
Basal shows the steepest decline of any subtype
(log-rank $p = 0$) across all 7 subtypes)*



*Receptor profile confirms Basal $\approx$
ER-/HER2*

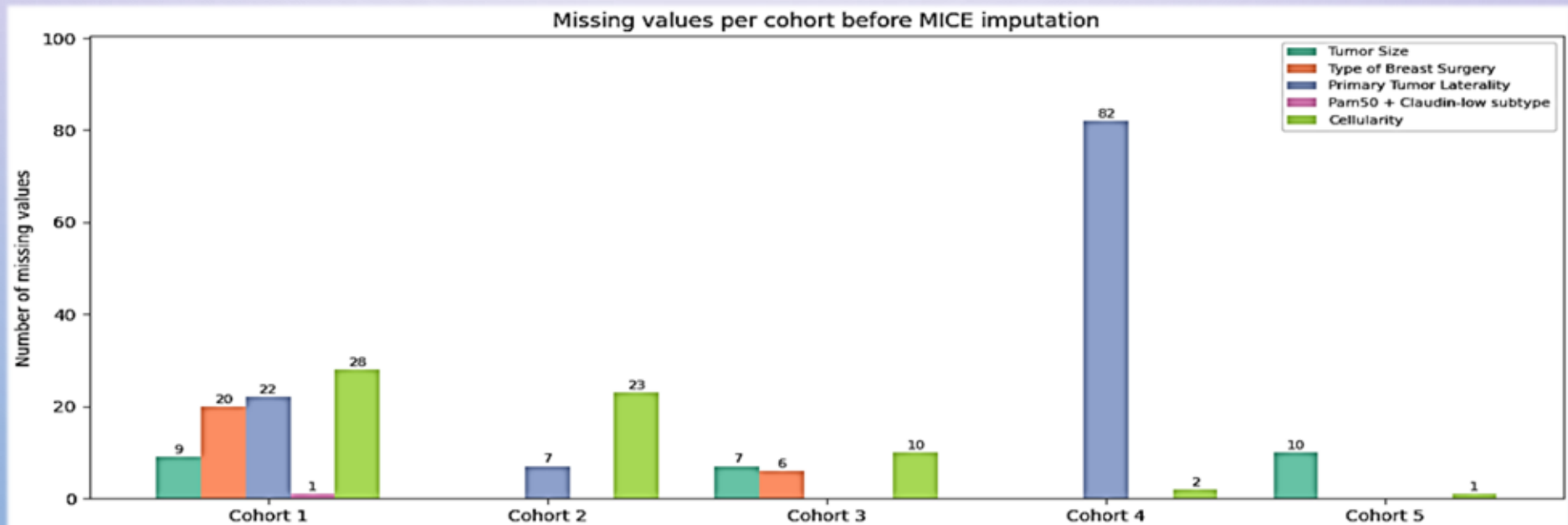


*Basal vs. rest confirms the gap
(pairwise $p = 0.31$, not significant
given small Basal n)*

HANDLING MISSING DATA WITH MICE

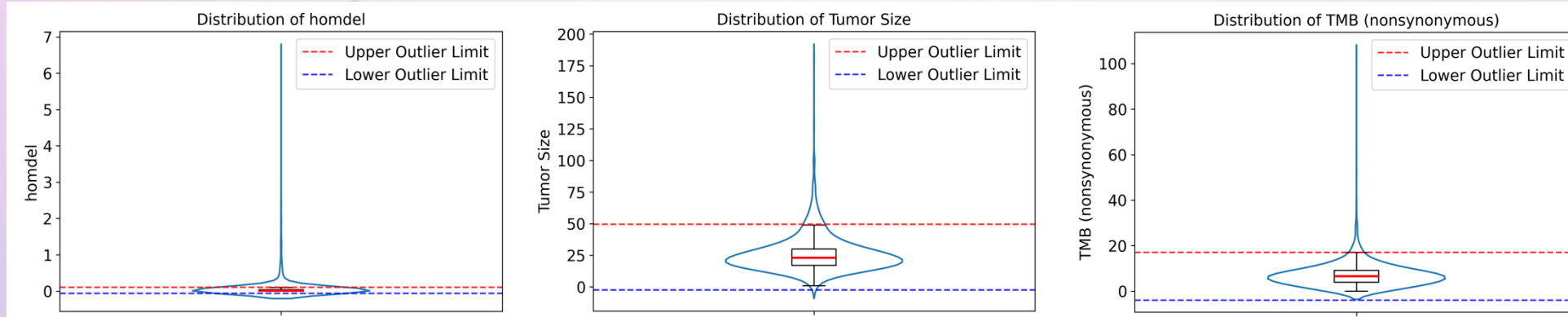
MULTIVARIATE IMPUTATION BY CHAINED EQUATIONS

- **Why MICE:** Missingness was scattered across all selected features numeric/categorical columns (Tumor Size, Cellularity, Tumor Laterality, etc.) — too much data loss to simply drop rows.
- **How it works:** MICE models each feature from the others, capturing realistic joint relationships rather than filling with a single mean/mode.
- **Per-cohort:** Imputation runs separately within each of the 5 cohorts, so no cohort's data leaks into another's imputed values.

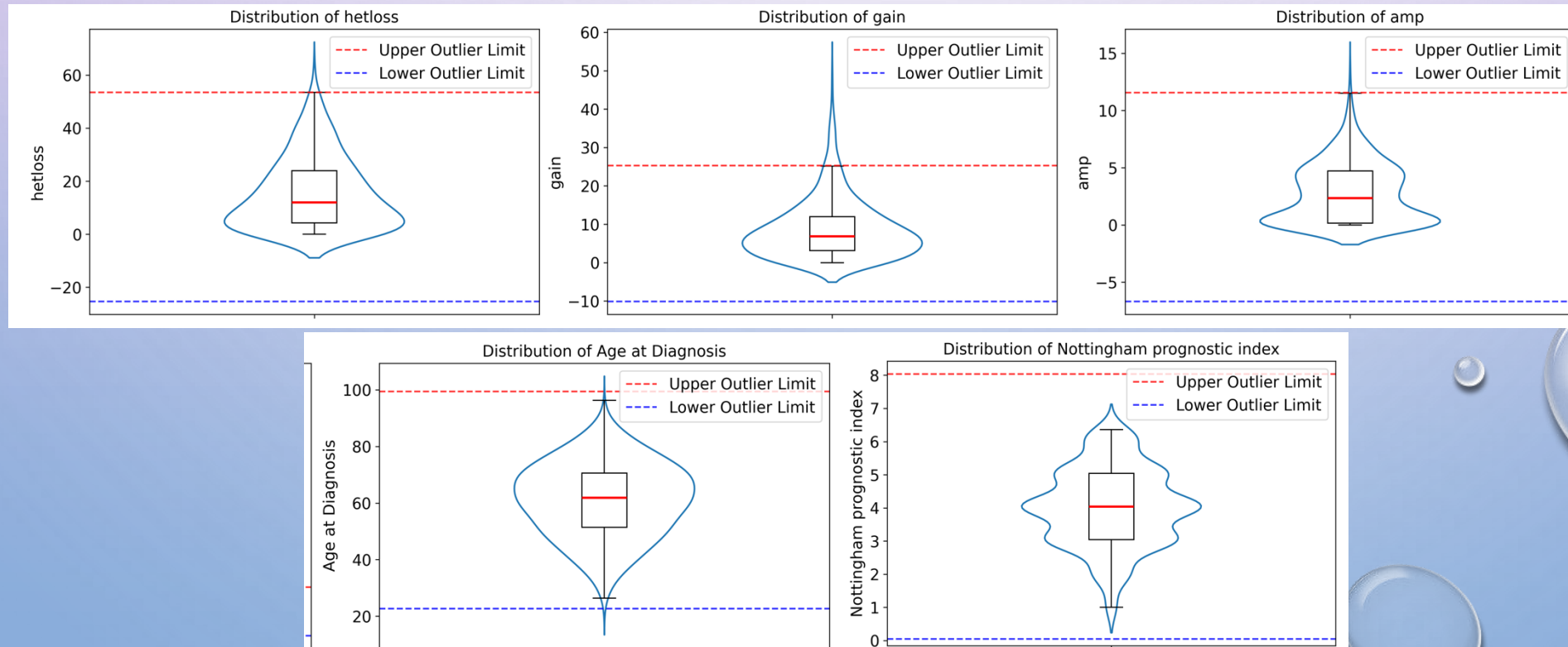


Data Integrity & Feature Audit

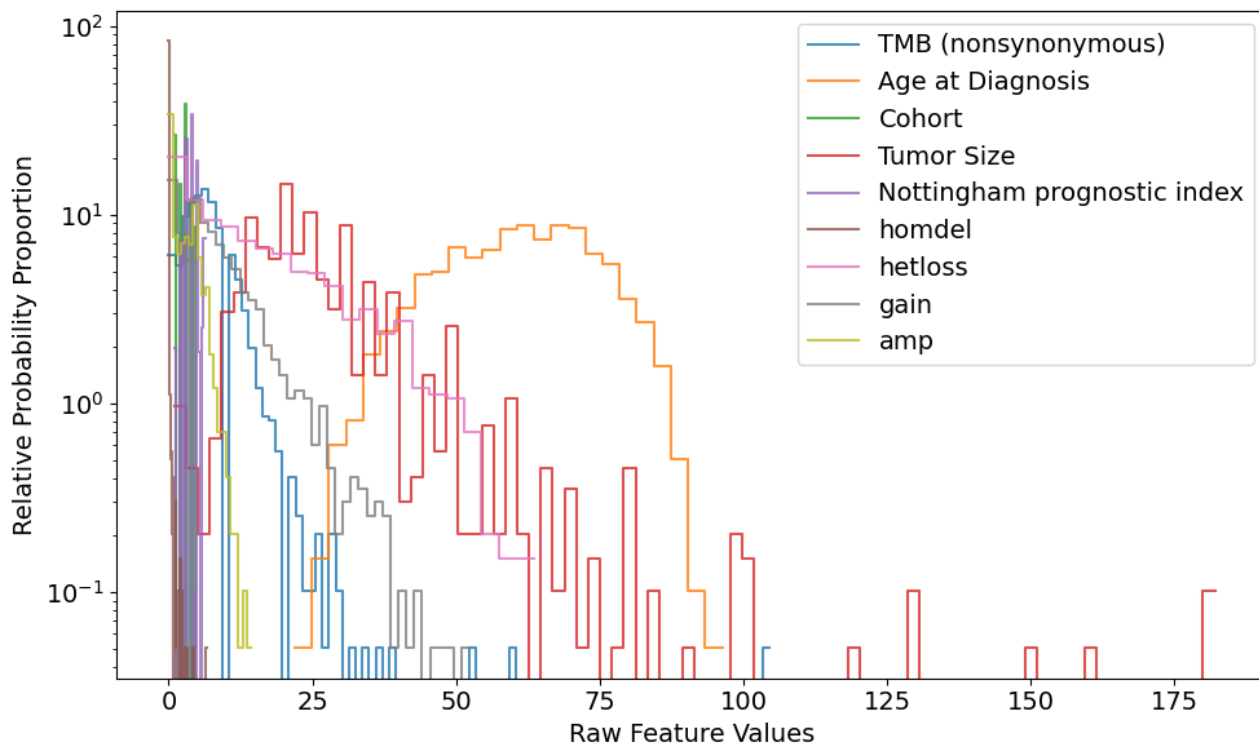
Right Skew



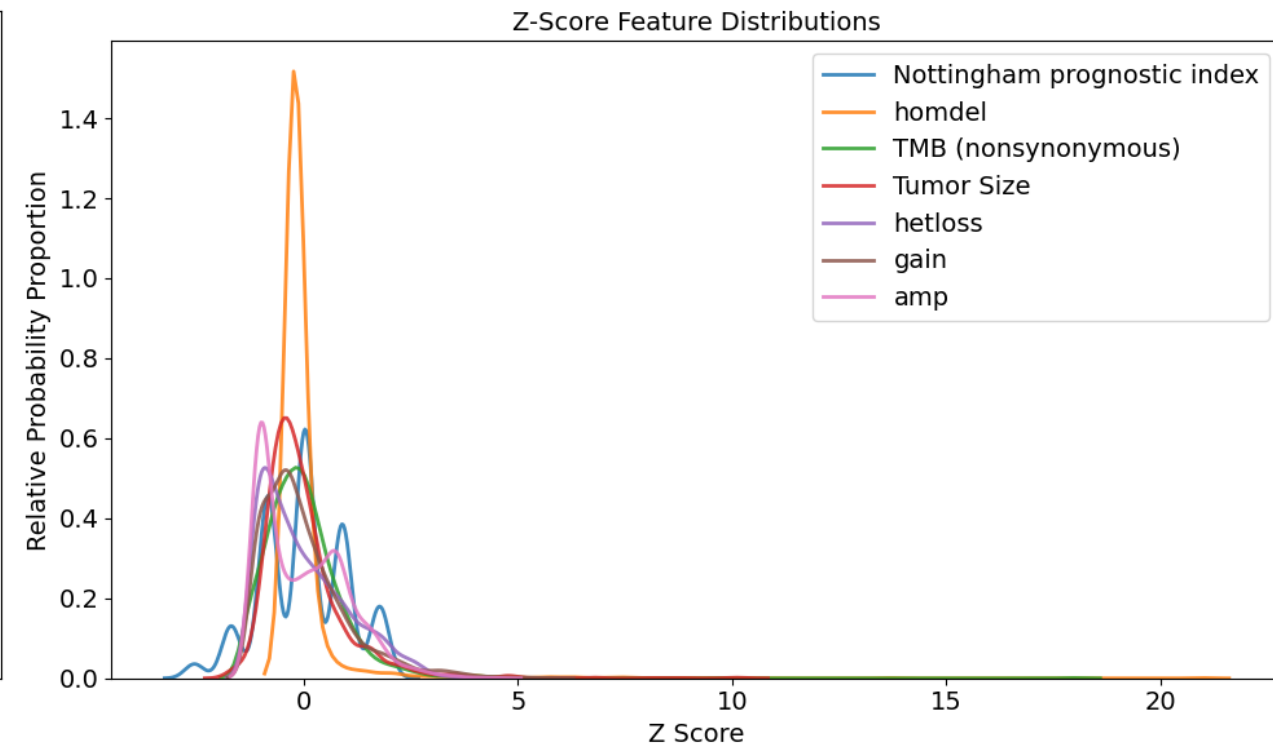
Normal Distributions



Is Cohort A Good Choice For CV?



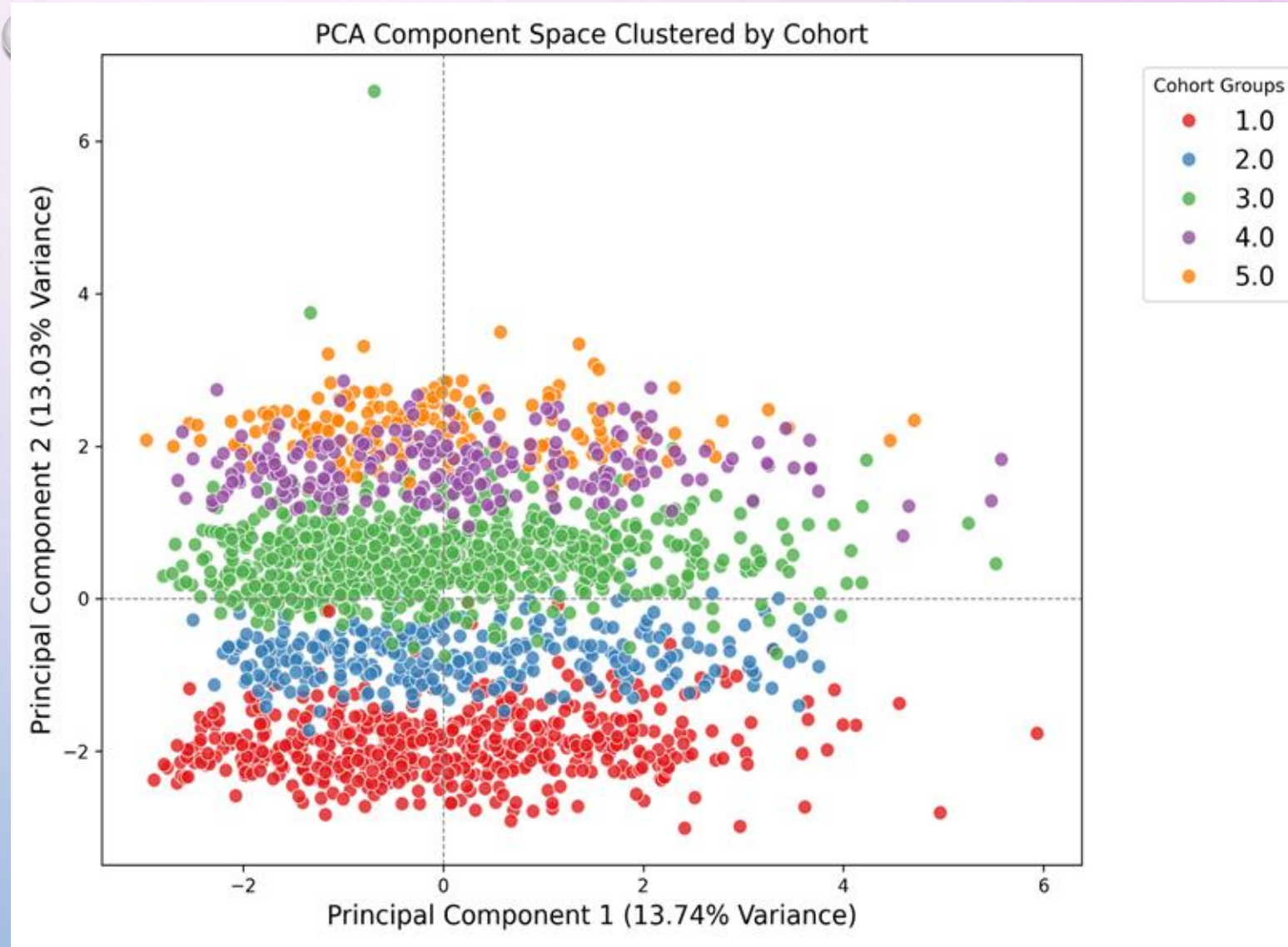
A



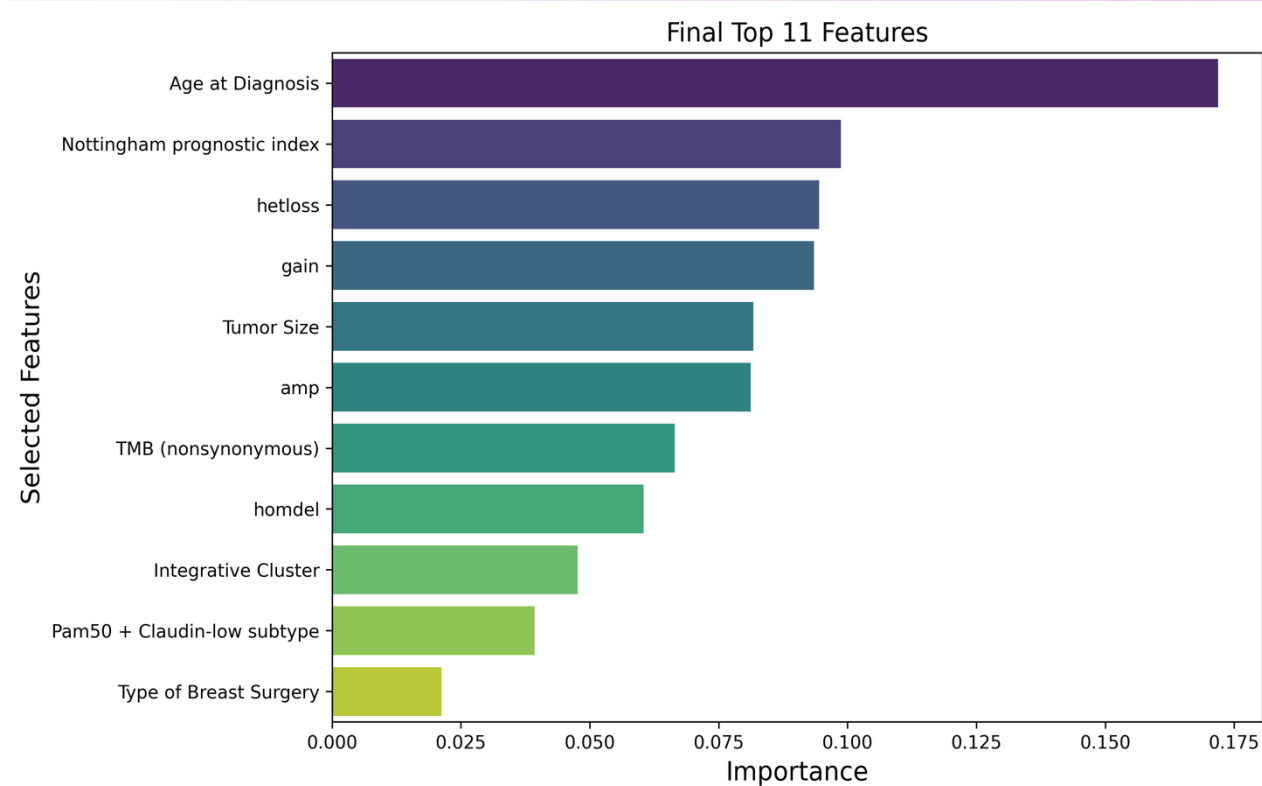
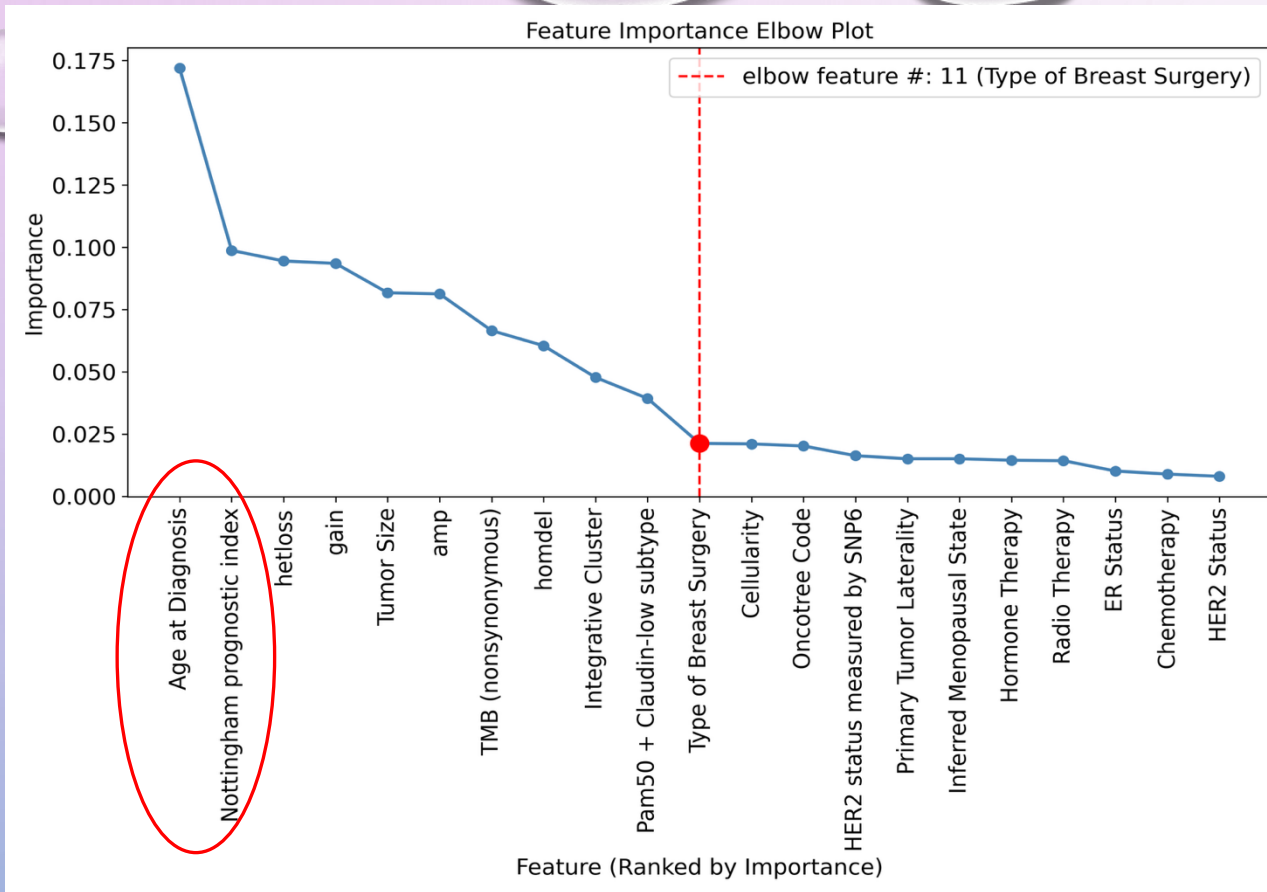
B

Is Cohort A Good Choice For CV?

- Principal Component Analysis showed non-overlapping cohorts (1–5) along PC2
- Inter-cohort separation refutes **random splitting**
- Utilized **cohort-based cross-validation** to maintain structural integrity.



RF-based Feature Selection & Elbow Optimization



Elbow Method Selection: Identifies **inflection point (red)** of importance, optimizing our feature space.

A

B

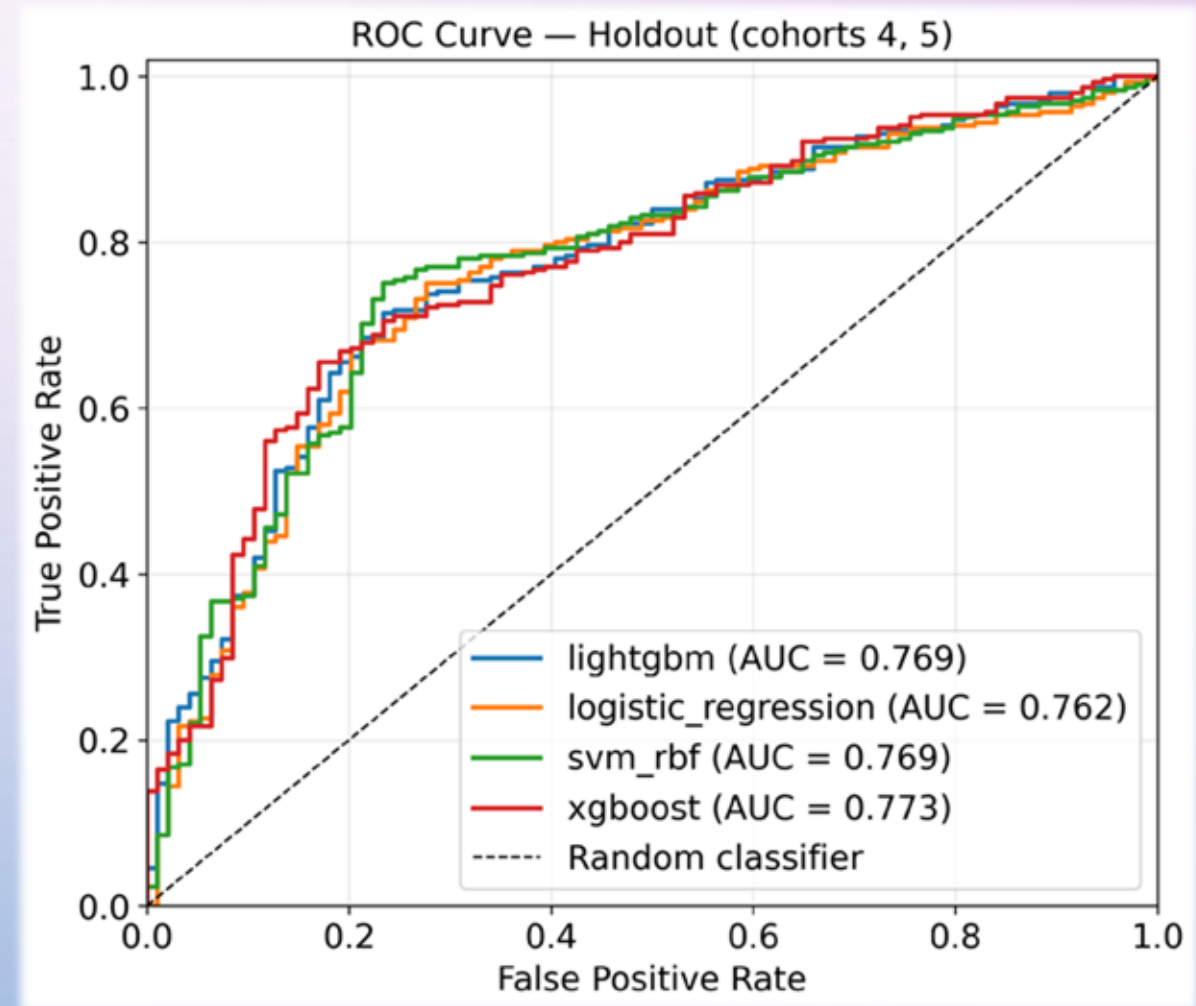
PREDICTING 5-YEAR SURVIVAL: MODEL COMPARISON

- **Cross-validation** : 1,518 patients (cohorts 1–3), **Test** : 399 patients (cohorts 4–5), **total** : **1,917**
- **Best ROC-AUC:** SVM (0.79).
- **Best minority-class performance:** XGBoost — highest MCC (0.40) and highest F1 on the non-survival ('No') class (0.56).
- **Overall:** all four models cluster tightly (ROC-AUC 0.76–0.79) — no single model dominates across every metric.

	ROC-AUC	PR-AUC	MCC	Precision survival		Recall survival		f1 survival	
				No	Yes	No	Yes	No	Yes
Logistic Regression	0.76	0.89	0.36	0.39	0.90	0.78	0.62	0.53	0.74
Light GBM	0.76	0.91	0.39	0.41	0.90	0.74	0.67	0.55	0.78
XGBoost	0.77	0.91	0.40	0.43	0.91	0.78	0.69	0.56	0.78
SVM	0.79	0.92	0.38	0.42	0.91	0.79	0.66	0.54	0.76

MODEL DISCRIMINATION: ROC CURVE

- **Ranking (holdout AUC):** XGBoost (0.773) $\approx$ SVM (0.769) $\approx$ LightGBM (0.769) $>$ Logistic Regression (0.762) — all four are close.
- **Shape:** All four sit well above the random baseline, with the most separation at low false-positive rates (FPR $<$ 0.2).



CONCLUSION & FUTURE DIRECTIONS

Project strengths

- Per-cohort mice imputation preserved all 1,917 patients without leaking data across the holdout split.
- Benchmarked four model families under the same leave-one-cohort-out validation for a fair comparison.
- Feature importance - the model relies on the same factors doctors already trust (age, tumor size, prognostic index) — but subtype and genomic data added real predictive value on top

Limitations & next steps

- Excluded censored patients (alive, <5yr follow-up) — a Cox/DeepSurv model would use them instead.
- Validated only within METABRIC — next step is an independent cohort like TCGA-BRCA.Clinical + copy-number features only — gene-expression signatures could add more.
- Decision threshold not yet tuned to the real cost of false negatives vs. false positives.